

Phosphate is a vascular toxin.

Elevated phosphate (P) levels are seen in advanced renal failure and, together with dysregulated calcium, parathyroid hormone and vitamin D levels, contribute to the complex of chronic kidney disease-mineral and bone disease (CKD-MBD). Converging evidence from in vitro, clinical and epidemiological studies suggest that increased P is associated with vascular calcification and mortality. When vessels are exposed to high P conditions in vitro, they develop apoptosis, convert to bone-like cells and develop extensive calcification. Clinical studies in children on dialysis show that high P is associated with increased vessel wall thickness, arterial stiffness and coronary calcification. Epidemiological studies in adult dialysis patients demonstrate a significant and independent association between raised P and mortality. Importantly, raised P is associated with cardiovascular changes even in pre-dialysis CKD, and also in subjects with normal renal function but high P. All P binders can effectively reduce serum P, and this decrease is linked to improved survival. Raised serum P triggers the release of fibroblast growth factor 23 (FGF-23), which has the beneficial effect of increasing P excretion in early CKD, but is increased several 1,000-fold in dialysis, and may be an independent cardiovascular risk factor. Both FGF-23 and its co-receptor Klotho may have direct effects on the vasculature leading to calcification. Fascinatingly, disturbances in FGF-23-Klotho and raised P have also been associated with premature aging. These data suggest that high P levels have adverse vascular effects and that maintaining the serum P levels in the normal range reduces cardiovascular risk and mortality.